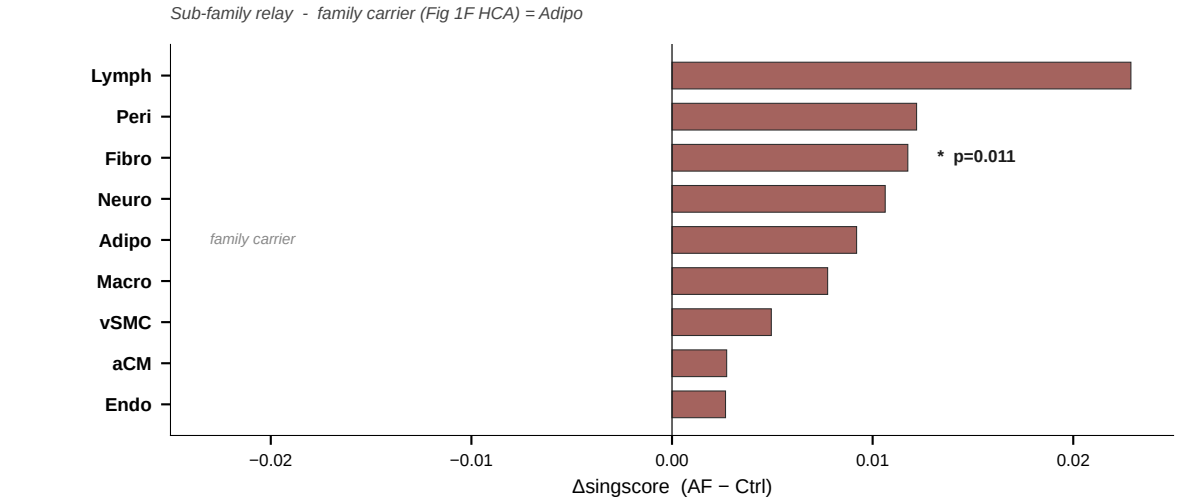
Fig 3f F4 BCAA catabolism (program-level)



Although F4 (4 metabolic programs) shows no family-level cellular signal when averaged,
its BCAA catabolism component reveals fibroblast as the AF-active site (p = 0.011) —
a sub-family relay from the family's adipocyte baseline carrier.
Consistent with the documented metabolic reprogramming of cardiac fibroblasts during AF-associated remodelling.

Note on Lymph: lymphocyte shows the largest ΔScore (+0.023) but does not reach nominal p < 0.05
(p = 0.14) because of higher within-group variance in the smaller-N lymphocyte sample (n = 10 Ctrl).
Fibroblast is therefore the only compartment passing the gene-level significance threshold here.

Color note: bars are colored by direction (RED = AF > Ctrl upshift, BLUE = AF < Ctrl downshift),
the same convention as Fig 3b, 3d and 3e. Color encodes direction (up/down), NOT cellular-fidelity
class (relay/extension/fidelity); the latter is shown by the badge color in each case-study panel.